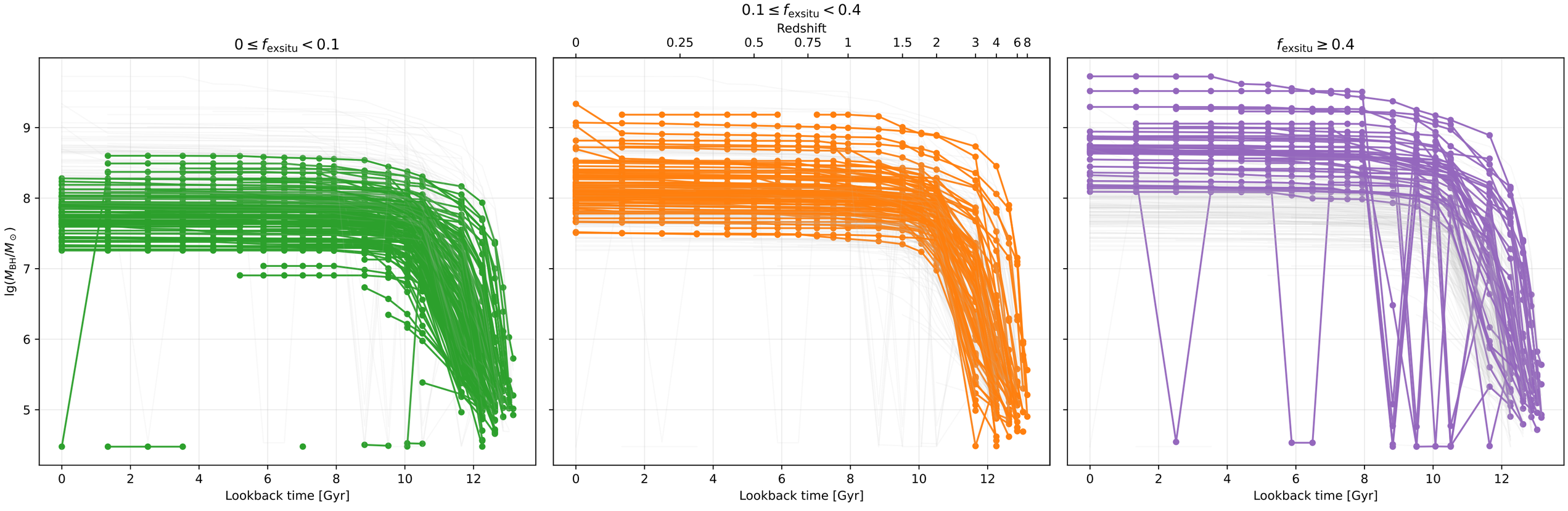

$0 \leq f_{\mathrm{exsitu}} < 0.1$
$0.1 \leq f_{\mathrm{exsitu}} < 0.4$
$f_{\mathrm{exsitu}} \geq 0.4$
Redshift
0
0.25
0.5
0.75
1
1.5
2
3
4
6
8
$\lg(M_{\mathrm{BH}}/M_{\odot})$
9
8
7
6
5
0
2
4
6
8
10
12
Lookback time [Gyr]